CSU San Marcos
Department of Mathematics
333 Twin Oaks Valley Rd.
San Marocs, CA 92078
`kayers@csusm.edu`

Education

Ph.D. in Mathematics, Iowa State University 2015
Dissertation: **Graph Determined Symbolic Dynamics and Hybrid Systems**.
Adviser: Dr. Wolfgang Kliemann

B.A. in Mathematics, Bowdoin College, Brunswick, Maine 2010
Honors Thesis: "Stochastic Perturbations of the Fitzhugh-Nagumo Equations"
Adviser: Dr. Mary Lou Zeeman.

Employment

Assistant Professor, California State University San Marcos	2021 - present
Assistant Professor, Carroll College	2019 - 2021
Visiting Assistant Professor, Pomona College	2016 - 2019

Honors and Awards

Project NExT Fellow, Peach '18 Cohort	2018-2019
Aggie Ho Research Award, Iowa State University	2015
Graduate College Research Excellence Award, Iowa State University	2014
Edward S. Hammond Prize, Mathematics Department, Bowdoin College	2010
Graduated with Honors, Mathematics Department, Bowdoin College	2010

Publications and Papers

9. W. Aitken, K. Ayers, H. Smith. "General Boyd-Lawton theorems with multivariable limits." Submitted, 2025.

8. K. Ayers, M. Kooiker. "A structural analysis of population graphs." Submitted, 2025.

7. K. Ayers, A. Radunskaya. 'Stability of the invariant distribution of the stochastic logistic map." Conditionally accepted in *Research in the Mathematics Sciences*, January 2025.

6. K. Ayers, D. Dmitrishin, A. Radunskaya, A. Stokolos. "Search for invariant sets of the generalized tent map." *Journal of Difference Equations and Applications*, 29(9-12), 1156-1183, 2022. https://doi.org 10.1080/10236198.2022.2119851

5. K. Ayers. "Chain Recurrence in Graph Determined Hybrid Systems". *Journal of Dynamics and Control Systems* (2019). https://doi.org/10.1007/s10883-019-09440-x

4. H. Cho, K. Ayers, L. de Pillis, Y-H. Kuo, J. Park, A. Radunskaya, R. Rockne. “Modeling acute myeloid leukemia in a continuum of differentiation states.” *Letters in Biomathematics*, 5(sup1): S69-S98, 2018.
3. K. Ayers. “Graph Determined Symbolic Dynamics and Hybrid Systems,” Ph.D. dissertations, Iowa State University: ProQuest/UMI, 2015
2. J. Ackerman, K. Ayers, E. Beltran, J. Bonet, D. Lu, T. Rudelius. ” A behavioral characterization of Markov chains and discrete time dynamical systems over directed graphs.” *Qualitative Theory of Dynamical Systems*, 13(1):161-180, 2014.
1. K. Ayers, X. Garcia, J. Kunze, T. Rudelius, A. Sanchez, S. Shao, E. Speranza. “Limit and Morse sets of deterministic hybrid systems.” *Qualitative Theory of Dynamical Systems*, 12(2):335-360, 2013.

Grants and Fellowships

NSF Improving Undergraduate Stem Education (IUSE) 2025-2028
Title: “Collaborative Research: Elevating the calculus standard of care through an innovative collaborative testbed”
CSUSM PI $173,098

AMS PUI Travel Grant 2023
Joint Mathematics Meetings $2100
Co-PI

AMS PUI Travel Grant 2022
Joint Mathematics Meetings - on hold $2,100

NIH Research Enhancement Award (R15) 2021-2024
Title: “The Architecture of Missing and Archaic Variation in Human Population Genomic Data”
PI $440,087

AIM Math Community 2020-present
“Little School Dynamics - A Dynamical Systems Community Projects for Researchers at PUIs”
Co-PI

Professional Presentations

“Evolutionary Dynamics, Population Graphs, and Ghost Populations” 2024
International Society for Mathematical Biology Conference, Seoul, Korea

“Stability of the invariant distribution of the stochastic logistic map” 2024
JMM AIM Special Session on Little School Dynamics: Cool Research Done at PUIs

“Stability of the invariant distribution of the stochastic logistic map” 2024
Invited Talk, University of Utah Applied Mathematics Seminar

“Stability of the invariant distribution of the stochastic logistic map” 2023
CSU Mathematical Sciences Conference

“A Chaotic Introduction” 2023
Route 78 Math Day

“Search for invariant sets of the generalized tent map” 2023
JMM AIM Special Session on Little School Dynamics: Cool Research Done at PUIs

“An Exploration of Sharkovskii’s Theorem” 2022
University of San Diego Invited Colloquium Talk

“Baby Sharkovskii” 2022
San Marcos Informal Mathematics In-Person Colloquium

“Selected Topics in Hybrid Systems” 2022
Invited Talk, Boston Computing Group

“Random Perturbations of Self-Regulating Biological Processes” 2022
JMM AWM Special Session on Women in Mathematical Biology

“Harris Irreducibility of the Stochastic Logistic Map” 2022
JMM AMS Special Session on Little School Dynamics: Cool Research done at PUIs

“Fixed Points, Stability, and Oscillations: How can we manipulate systems?” 2022
Invited Colloquium Talk, Cal State Long Beach

“Fixed Points, Stability, and Oscillations: How can we manipulate systems?” 2022
Invited Colloquium Talk, Skidmore College

“Little School Dynamics - Research at PUIs” 2022
Invited Talk, SIMIODE Expo

“Random Perturbations of Self-Regulating Biological Processes” 2021
UCSD NITMB Colloquium

“Fixed Points, Stability, and Oscillations: How can we manipulate systems?” 2021
San Marcos Mathematics Online Colloquium

“Stochastic Logistic Maps and Invariant Distributions” 2021
Invited Graduate Seminar, University of Georgia

“Stochastic Perturbations of Unimodal Maps” 2021
Invited Colloquium Talk, Cal State San Marcos

“Perceiving Patterns and Chaos” 2021
Cal State San Marcos Student Seminar

“Random Perturbations of Unimodal Maps” 2021
AIM Math Community - Little School Dynamics Invited Talk

“Faculty Dynamics Research Opportunities - Primarily Undergraduate Institutions” 2021
Invited Panel, SIMIODE Expo

“Stabilization of Quasi-Periodic Orbits of the Random Logistic Map” 2020
Invited Talk, Women in Math in Southern California Conference

“Chain Recurrence in Graph Determined Hybrid Systems” 2020
MAA Contributed Paper Session, Joint Mathematics Meetings 2020

“A Skew Product Model for Hybrid Dynamical Systems” 2019
Invited Colloquium Talk, University of Montana

“Stochastic Perturbations of the Logistic Map.” 2019
MAA Invited Paper Session, MathFest 2019

“Perceiving Patterns and Chaos” 2019
Invited Colloquium Talk, Carroll College

“A Skew Product Model for Hybrid Dynamical Systems” 2018
Invited Colloquium Talk, California Polytechnic Institute, Pomona

“A Chaotic Introduction” 2018
Claremont Math Weekend, Claremont, CA

“Perceiving Patterns and Chaos” 2017
Invited Colloquium Talk, Colorado College

“Perceiving Patterns and Chaos” 2017
Invited Colloquium Talk, Lafayette University

“Perceiving Patterns and Chaos” 2017
Invited Colloquium Talk, Agnes Scott College

“Skew Product Flows and Hybrid Systems” 2017
Applied Mathematics Seminar, Pomona College, CA

“Skew Product Flows and Hybrid Systems” 2017
Women in Math in Southern California Conference, Los Angeles, CA

“Symbolic Dynamics and Deterministic Hybrid Systems” (poster) 2015
Joint Mathematics Meetings, San Antonio, TX

“Symbolic Dynamics and Deterministic Hybrid Systems” 2013
Mathematics Colloquium, University of Augsburg, Augsburg, Germany

Other Presentations

“Panel Industry and Research in CSTEM” 2022
Associated Students Inc. Invited Panelist

“STEM and Twitter” 2022
CSUSM oSTEM Chapter Presentation

“The Math of Climate Science” 2022
Climate Teach in, CSUSM, CA

“Queering Math” 2018
Mathematics Community Seminar, Department of Mathematics, Pomona College, CA

“Queering Math” 2017
FoRCES Reading Group, Pomona College, CA

Professional Workshops

Harvard Master Class Teaching Math Modeling for Life Science — Summer 2023
Teaching Math Modeling for Life Science — Funded Senior Participant

CSUSM Faculty Learning Community Contemplative Pedagogy for Social Justice — Fall 2021
Contemplative Pedagogy for Social Justice — Accepted Participant

ICERM Virtual Workshop — June 2021
Advanced Workshop in Data Science for Mathematical Sciences Faculty — Accepted Participant

MAA Virtual Workshop — April 2021
How and Why Should Sustainability Be Part of What We Teach? — Accepted Participant

Joint Mathematics Meetings, Denver, CO — January 2020
AWM Workshop: Moving Toward Action — Funded Participant

Joint Mathematics Meetings, Baltimore, MD — January 2019
MAA Minicourse: Start Teaching Statistics using R and RStudio — Funded Participant

Mathematical Association of America MathFest 2018, Denver CO — August 2018
Project NExT Participant — Funded Participant

Iowa State University — Summer 2009
NSF-Sponsored REU Program — Funded Participant

CSUSM Professional Service

Student Affairs Committee — 2024-present
CSTEM Representative

STEM Strategic Planning Committee on AI Member	2024-2025
Academic Senate Senator representing CSTEM	2023-2025
San Marocs Informal Mathematics In-person Colloquium (SMIMIC) Co-Coordinator	2023-present
Reid Lecture Committee	2022-present
Math 150 Course Coordinator	2024 - present
Math 132 Community of Practice Tenure-Track Faculty Participant	Fall 2023
Faculty Center Teaching and Learning Group Group member	2022-present
AWM Student Chapter Faculty Advisor	2021 - present
oSTEM Chapter Faculty Advisor	2021 - present
Committee to create a new life sciences calculus Lead Committee Member	Summer 2022
Cal-Bridge Program Cal State Mentor	2022 - 2023
2021-2022 Math Department Search, Tenure Track Line Search Committee Member	2021-2022
Math 132 Course Coordinator	2021-2022

Other Professional Service

AWM Executive Committee 2024 - present
Social Media Coordinator

MAA SoCal-Nevada Section 2023-2026
Vice Section Chair/Section Chair/Past Section Chair

AMS Special Session, JMM 2024 2024
Organizer

SIAM Conference on Applications of Dynamical Systems 2023
Mini-Symposium Organizer

AMS Special Session, JMM 2023 2023
Organizer

CSU JMM, CSU Northridge 2022
Plenary Talk Chair

Johns Hopkins University Press, Kyne Santos Untitled Book 2022
Book Reviewer

CSU JMM Scientific Committee 2022
Committee Member

AMS Special Session, JMM 2022 2022
Organizer

AMS Math Reviews 2020-present
Reviewer

AWM Media Committee 2019-present
Member

Pomona Academy for Youth Success (PAYS) 2017 - 2021
Math Instructor

Carroll College Safe Zone Training 2021
Co-Facilitator

AWM The First Fifty Years: Reminiscences, Participant History, and Visions for the Future 2020
Article Referee

Carroll College Math Debate 2020
Debater: Who is the most influential mathematician who died young? (Debate Winner Representing Srinivasa Ramanujan)

AMS Social, JMM 2020 2020
Performer, Violinist

JMM MAA Workshop: Identifying and Managing Microaggressions in the Academic Setting 2020
Small group discussion facilitator

EDGE (Enhancing Diversity in Graduate Education) 2019
MATLAB Mini Course Instructor

Project NExT Session on Math Education at 2019 JMM 2019
Facilitator

AWM Strategic Task Force on Diversity and Inclusion 2018
Committee Member

Pomona College 1st Annual Sonia Kovalevsky Day 2018
Organizer

Nebraska Conference for Undergraduate Women in Mathematics 2013
Invited Panelist, Life in Graduate School

Bowdoin College Student Math Representative 2009-2010

Media Appearances

LGBTTech 2022
PATHS In-Person Networking Event Invited Panelist

My Favorite Theorem 2022
Podcast Guest

LGBTTech 2022
PATHS Program Interview

Undergraduate Student Projects

Haley Lorenz and Carmen Gutierrez — Summer 2023
"Visualizing the Invariant Distribution of the Stochastic Logistic Map"

Elisa Zepeda — 2020-2021
"Random Forests for Prenatal Maternal Data"

Eisen Ipac — 2018-2019
"Population dynamics in crime dynamics modeling"

Benjamin Gregory — 2017-2018
"Symbolic dynamics of Smale's horseshoe"

Abigail Gardner — 2016-2017
"Variation of parameters in crime dynamics modeling"

Master's Student Thesis Projects

Alireza Pakravan — 2021-2022
"Solving PDEs Using Neural Networks"

CSUSM Courses Taught

Fall 2025:
Math 350 - Foundations for Theoretical Mathematics
Math 362 - Differential Equations
Math 378 - Number Systems
Spring 2025:
Math 150 - Calculus for Life Sciences (2 Sections)
Math 378 - Number Systems
Fall 2024:
Math 150 - Calculus for Life Sciences
Math 378 - Number Systems
Spring 2024:
Math 350 - Foundations for Theoretical Mathematics
Math 364 - Intermediate Linear Algebra
Fall 2023:
Math 350 - Foundations for Theoretical Mathematics
Math 430 - Real Analysis
Spring 2023:
Math 350 - Foundations for Theoretical Mathematics

Math 430 - Real Analysis
Fall 2022:
Math 350 - Foundations for Theoretical Mathematics
Math 448 - Mathematical Methods in Biology
Spring 2022:
Math 132 - Survey of Calculus
Math 362 - Differential Equations
Fall 2021:
Math 132 - Survey of Calculus
Math 346 - Mathematical Methods of Physics

Courses Taught (Other Institutions)

Carroll College:
Math 121 – Differential Calculus (Fall 2019, Fall 2020)
Math 122 – Integral Calculus (Spring 2020, Spring 2021)
Math 207 – Introduction to Statistics (Fall 2019, Spring 2020, Spring 2021)
Math 233 – Multivariable Calculus (Fall 2019, Fall 2020)
Math 306 – Real Analysis (Spring 2020, Spring 2021)
Math 401 – Abstract Algebra and Modern Geometry (Fall 2020)
Pomona College:
Math 30 – Differential Calculus (Spring 2018)
Math 31 – Integral Calculus (2 sections, Fall 2016, Fall 2018)
Math 32 – Multivariate Calculus (2 sections, Spring 2017, Fall 2017)
Math 58 – Introduction to Statistics (Spring 2019)
Math 60 – Linear Algebra (Fall 2017, Spring 2018)
Math 101 – Introduction to Real Analysis (Fall 2018, Spring 2019)
Math 102 – Differential Equations with Modeling (Fall 2016)

Professional Memberships

American Mathematical Society
Mathematical Association of America
Association for Women in Mathematics
National Association of Mathematicians
SACNAS
Spectra

Computer Skills

Python
R
LaTeX
MATLAB
Mathematica
Git